

3.3.3 Derivatives Worksheets

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions, using the Quotient Rule when possible.

a) $y = \frac{x^2 + 4x + 3}{\sqrt{x}}$

If we simplify first then we get $y = x^{3/2} + 4x^{1/2} + 3x^{-1/2}$ and therefore

$$\frac{dy}{dx} = \frac{3}{2}x^{1/2} + 2x^{-1/2} - \frac{3}{2}x^{-3/2}$$

If we start by directly using the Quotient Rule we get

$$\frac{dy}{dx} = \frac{\sqrt{x}(2x+4) - \frac{1}{2\sqrt{x}}(x^2 + 4x + 3)}{(\sqrt{x})^2}$$

**Remember that simplifying your final derivative is not required on Exam 2.*

Note that both answers are algebraically equivalent.

b) $h(t) = \frac{1+2t}{3-4t}$

$$\frac{dh}{dt} = \frac{(3-4t)(2) - (-4)(1+2t)}{(3-4t)^2} = \frac{6-8t+4+8t}{(3-4t)^2} = \frac{10}{(3-4t)^2}$$

**Remember that simplifying your final derivative is not required on Exam 2.*

c) $g(u) = \frac{u+1}{u^3+u-2}$

$$\frac{dg}{du} = \frac{(u^3+u-2)(1) - (3u^2+1)(u+1)}{(u^3+u-2)^2} = \frac{u^3+u-2-3u^3-3u^2-u-1}{(u^3+u-2)^2} = \frac{-3-2u^3-3u^2}{(u^3+u-2)^2}$$

**Remember that simplifying your final derivative is not required on Exam 2.*

d) $y = \frac{t}{(t-1)^2}$

We first rewrite the function as $y = \frac{t}{t^2 - 2t + 1}$

$$\frac{dy}{dt} = \frac{(t^2 - 2t + 1)(1) - (2t - 2)(t)}{(t^2 - 2t + 1)^2} = \frac{t^2 - 2t + 1 - 2t^2 + 2t}{(t-1)^4} = \frac{1 - t^2}{(t-1)^4} = \frac{-(1+t)(t-1)}{(t-1)^4} = \frac{-(1+t)}{(t-1)^3}$$

**Remember that simplifying your final derivative is not required on Exam 2.*

e) $w(r) = \frac{r^2 + \sqrt{r}}{2e^r + \pi}$

$$\frac{dw}{dr} = \frac{(2e^r + \pi)\left(2r + \frac{1}{2\sqrt{r}}\right) - (2e^r)(r^2 + \sqrt{r})}{(2e^r + \pi)^2}$$

**Remember that simplifying your final derivative is not required on Exam 2.*

f) $A(s) = \frac{s^2 + 2}{s^4 - 3s^2 + \sin(1)}$

$$\frac{dA}{ds} = \frac{(s^4 - 3s^2 + \sin(1))(2s) - (4s^3 - 6s)(s^2 + 2)}{(s^4 - 3s^2 + \sin(1))^2}$$

**Remember that simplifying your final derivative is not required on Exam 2.*

2. Find the equation of the tangent line to the curve at the given value of x .

a) $y = \frac{3x+1}{x^2+1}, x=1$

We calculate that $y' = \frac{(x^2+1)(3) - (2x)(3x+1)}{(x^2+1)^2}$. We have

$$\text{Point: } (1, 2) \text{ \& Slope: } m = y'(1) = \frac{(2)(3) - (2)(4)}{(2)^2} = \frac{-2}{4} = -\frac{1}{2}$$

Therefore, the equation of the tangent line is $y - 2 = -\frac{1}{2}(x - 1) \rightarrow y = -\frac{1}{2}x + \frac{5}{2}$.

b) $y = \frac{\sqrt{x}}{x+1}, x = 4$

We calculate that $y' = \frac{(x+1)\frac{1}{2\sqrt{x}} - (1)\sqrt{x}}{(x+1)^2}$. We have

Point: $\left(4, \frac{2}{5}\right)$ & Slope: $m = y'(4) = \frac{(5)\left(\frac{1}{4}\right) - (1)(2)}{(5)^2} = \frac{\frac{5}{4} - 2}{25} = \frac{-\frac{3}{4}}{25} = -\frac{3}{100} = -0.03$

Therefore, the equation of the tangent line is $y - 0.4 = -0.03(x - 4) \rightarrow y = -0.03x + 0.52$.

3. Find the equations of the tangent lines to the curve $y = \frac{x-1}{x+1}$ that are parallel to the line $x - 2y = 2$.

We calculate that $y' = \frac{(x+1)(1) - (1)(x-1)}{(x+1)^2} = \frac{2}{(x+1)^2}$.

We see that the line we want tangent lines parallel to is $x - 2y = 2 \rightarrow y = \frac{1}{2}x - 1$ which clearly has slope $\frac{1}{2}$.

We want to find a value of x at which $\frac{2}{(x+1)^2} = \frac{1}{2}$.

$$\frac{2}{(x+1)^2} = \frac{1}{2} \rightarrow 4 = (x+1)^2 \rightarrow \pm 2 = x+1 \rightarrow x = \pm 2 - 1 \rightarrow x = 1 \text{ or } x = -3$$

Therefore, we have tangent lines with slope $\frac{1}{2}$ at the points $(1, 0)$ & $(-3, -2)$.

So equations of the tangent lines are

$$y - 0 = \frac{1}{2}(x - 1) \rightarrow y = \frac{1}{2}x - \frac{1}{2} \text{ at } (1, 0) \text{ and}$$

$$y - (-2) = \frac{1}{2}(x - (-3)) \rightarrow y = \frac{1}{2}x - \frac{1}{2} \text{ at } (-3, -2)$$

4. Determine the value of $h'(2)$, given that $f(2) = -3$, $g(2) = 4$, $f'(2) = -2$, & $g'(2) = 7$.

a) $h(x) = \frac{f(x)}{2g(x) + x^2}$

First note that $h'(x) = \frac{(2g(x) + x^2)f'(x) - (2g'(x) + 2x)f(x)}{(2g(x) + x^2)^2}$.

Therefore,

$$\begin{aligned} h'(2) &= \frac{(2g(2) + (2)^2)f'(2) - (2g'(2) + 2(2))f(2)}{(2g(2) + (2)^2)^2} = \frac{(2(4) + 4)(-2) - (2(7) + 4)(-3)}{(2(4) + 4)^2} \\ &= \frac{(12)(-2) - (18)(-3)}{(12)^2} = \frac{-24 + 54}{144} = \frac{30}{144} = \frac{5}{24} \end{aligned}$$

b) $h(x) = \frac{g(x) + 1}{2 + f(x)}$

First note that $h'(x) = \frac{(2 + f(x))g'(x) - f'(x)(g(x) + 1)}{(2 + f(x))^2}$.

Therefore,

$$\begin{aligned} h'(2) &= \frac{(2 + f(2))g'(2) - f'(2)(g(2) + 1)}{(2 + f(2))^2} = \frac{(2 + (-3))(7) - (-2)(4 + 1)}{(2 + (-3))^2} \\ &= \frac{(-1)(7) - (-2)(5)}{(-1)^2} = 3 \end{aligned}$$

5. If f is a differentiable function, find an expression for the derivative of the function $y = \frac{1 + xf(x)}{\sqrt{x}}$.

We calculate that

$$y' = \frac{\sqrt{x}[0 + (f(x) + xf'(x))] - \frac{1}{2\sqrt{x}}(1 + xf(x))}{(\sqrt{x})^2}.$$

**Remember that simplifying your final derivative is not required on Exam 2.*

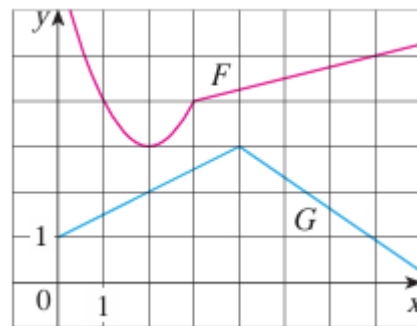
6. Let $P(x) = F(x)G(x)$ and $Q(x) = \frac{F(x)}{G(x)}$, where F and G are the functions whose graphs are shown below.

a) Find $P'(2)$

We know that $P'(x) = F'(x)G(x) + F(x)G'(x)$.

Therefore, $P'(2) = F'(2)G(2) + F(2)G'(2)$. From the graph we can determine each of these values and find that

$$P'(2) = F'(2)G(2) + F(2)G'(2) = (0)(2) + (3)\left(\frac{1}{2}\right) = \frac{3}{2}.$$



b) Find $Q'(7)$

We know that $Q'(x) = \frac{F'(x)G(x) - F(x)G'(x)}{G(x)^2}$.

Therefore, $Q'(7) = \frac{F'(7)G(7) - F(7)G'(7)}{G(7)^2}$. From the graph we can determine each of these values and find that

$$Q'(7) = \frac{F'(7)G(7) - F(7)G'(7)}{G(7)^2} = \frac{\left(\frac{1}{4}\right)(1) - (5)\left(-\frac{2}{3}\right)}{(1)^2} = \frac{1}{4} + \frac{10}{3} = \frac{3}{12} + \frac{40}{12} = \frac{43}{12}.$$